

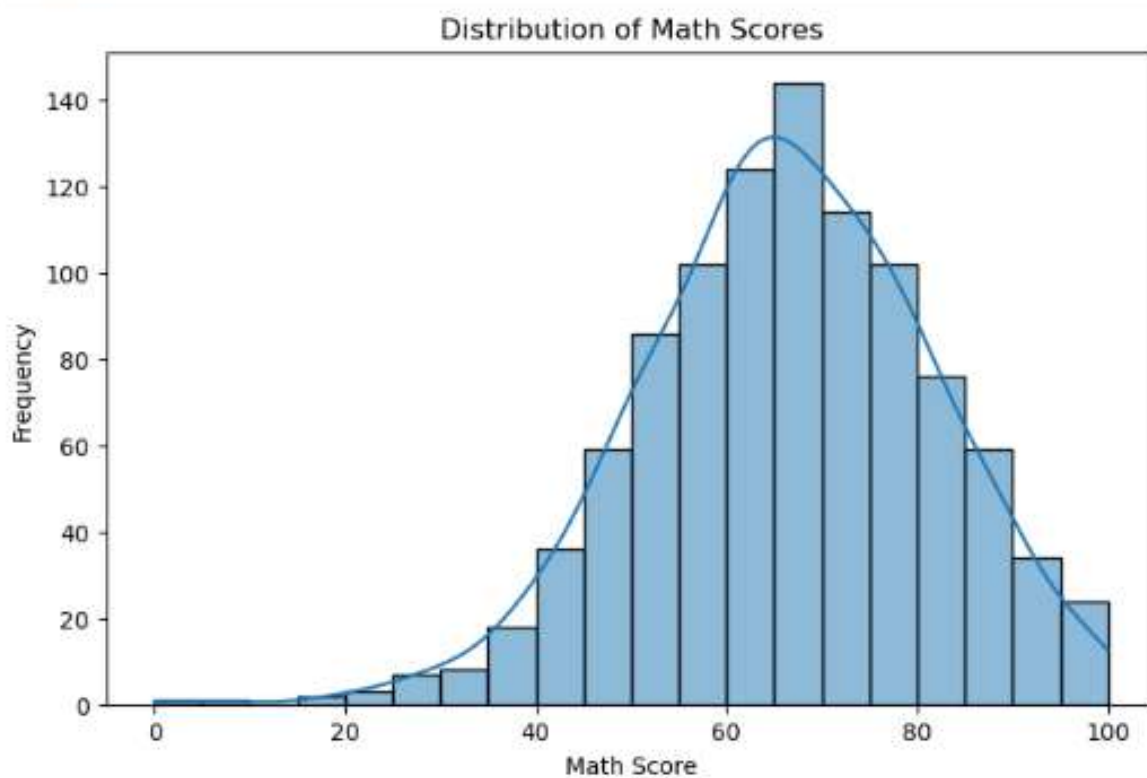
1. Distribution of Math Scores ¶

```
19]: plt.figure(figsize=(8,5))

sns.histplot(
    df["math score"],
    bins=20,
    kde=True
)

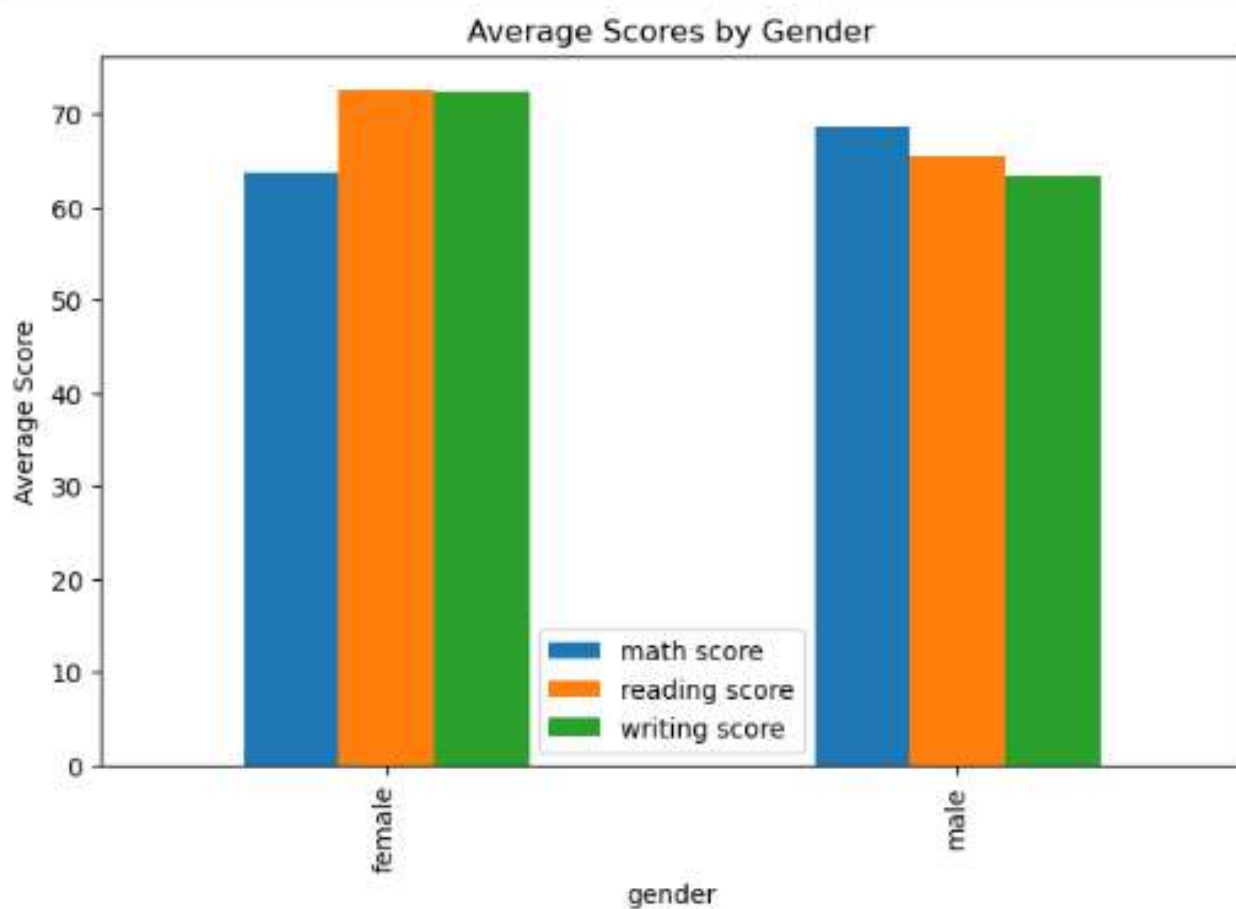
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")

plt.show()
```



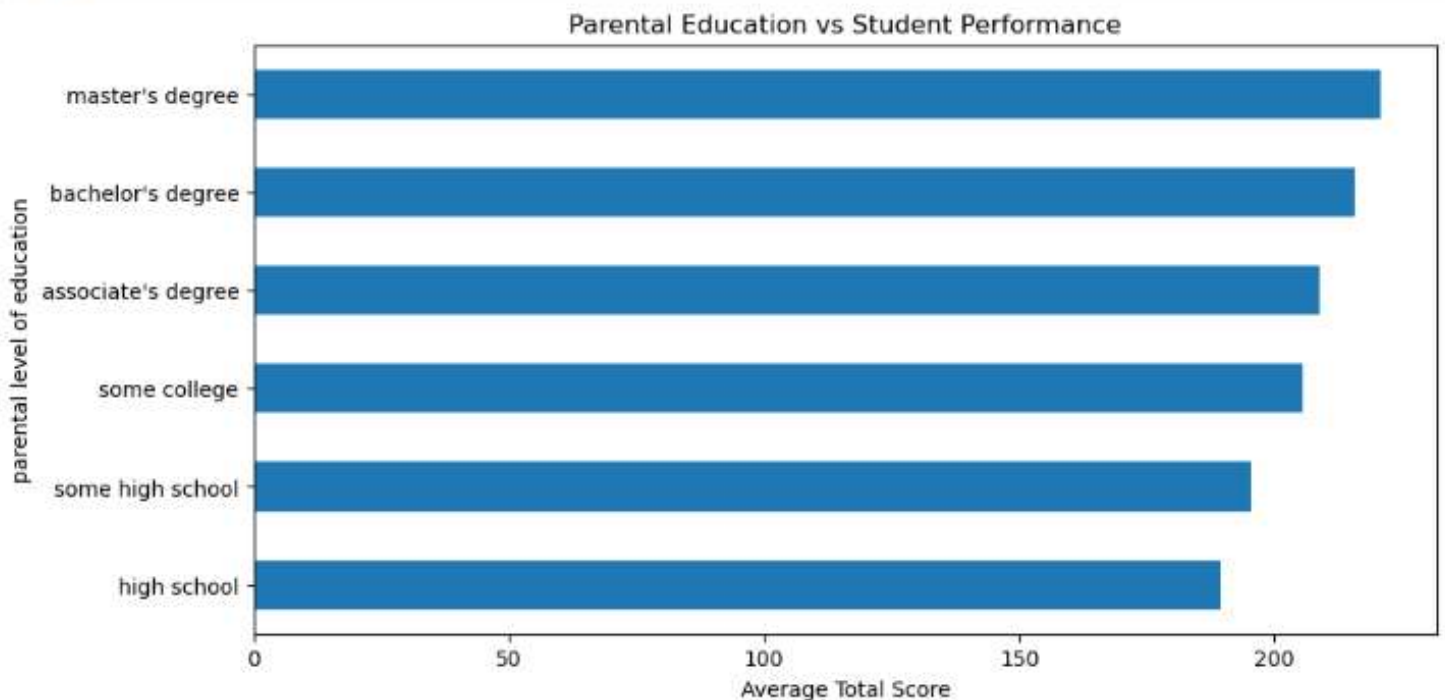
2. Average Scores by Gender

```
[20]: gender_scores = df.groupby("gender")[\n      ["math score",\n       "reading score",\n       "writing score"]\n      ].mean()\n\ngender_scores.plot(\n    kind="bar",\n    figsize=(8,5)\n)\n\nplt.title("Average Scores by Gender")\nplt.ylabel("Average Score")\n\nplt.show()
```



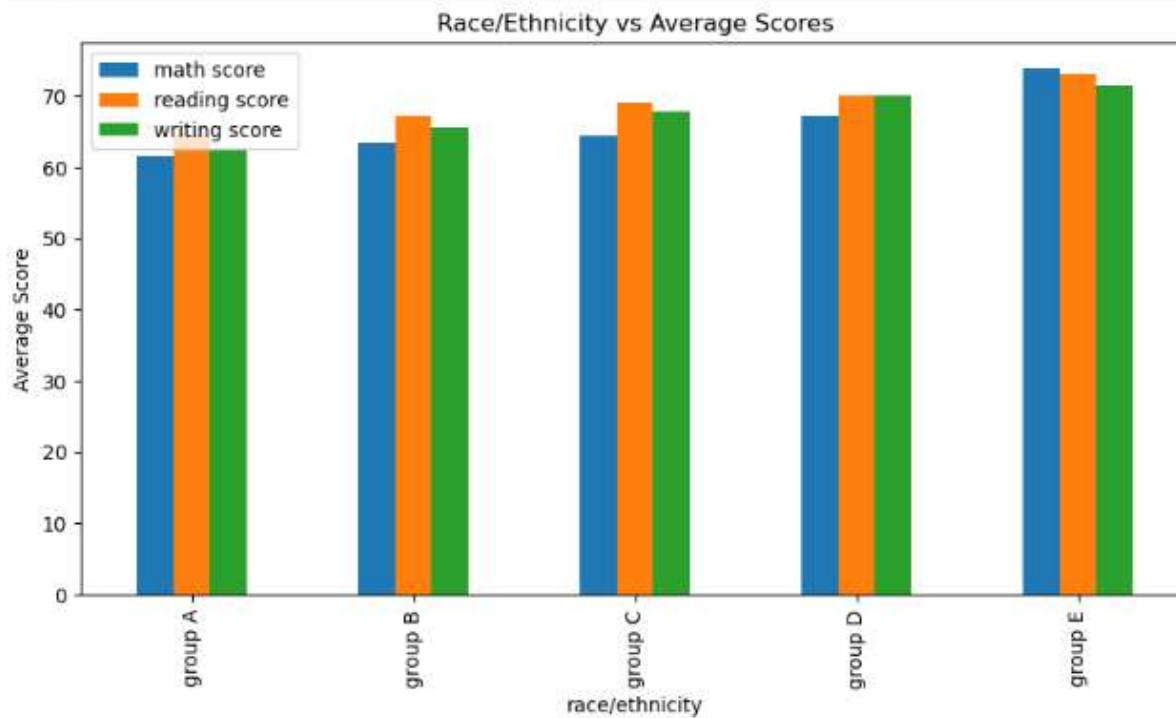
3. Parental Education vs Student Performance

```
[21]: education_scores = df.groupby(  
      "parental level of education"  
    )[  
      "total_score"  
    ].mean().sort_values()  
  
[22]: plt.figure(figsize=(10,5))  
  
education_scores.plot(  
    kind="barh"  
)  
  
plt.title(  
    "Parental Education vs Student Performance"  
)  
  
plt.xlabel("Average Total Score")  
  
plt.show()
```



4. Race/Ethnicity vs Average Scores

```
23]: race_scores = df.groupby(  
    "race/ethnicity"  
)[  
    ["math score",  
     "reading score",  
     "writing score"]  
].mean()  
  
24]: race_scores.plot(  
    kind="bar",  
    figsize=(10,5)  
)  
  
plt.title(  
    "Race/Ethnicity vs Average Scores"  
)  
  
plt.ylabel("Average Score")  
  
plt.show()
```



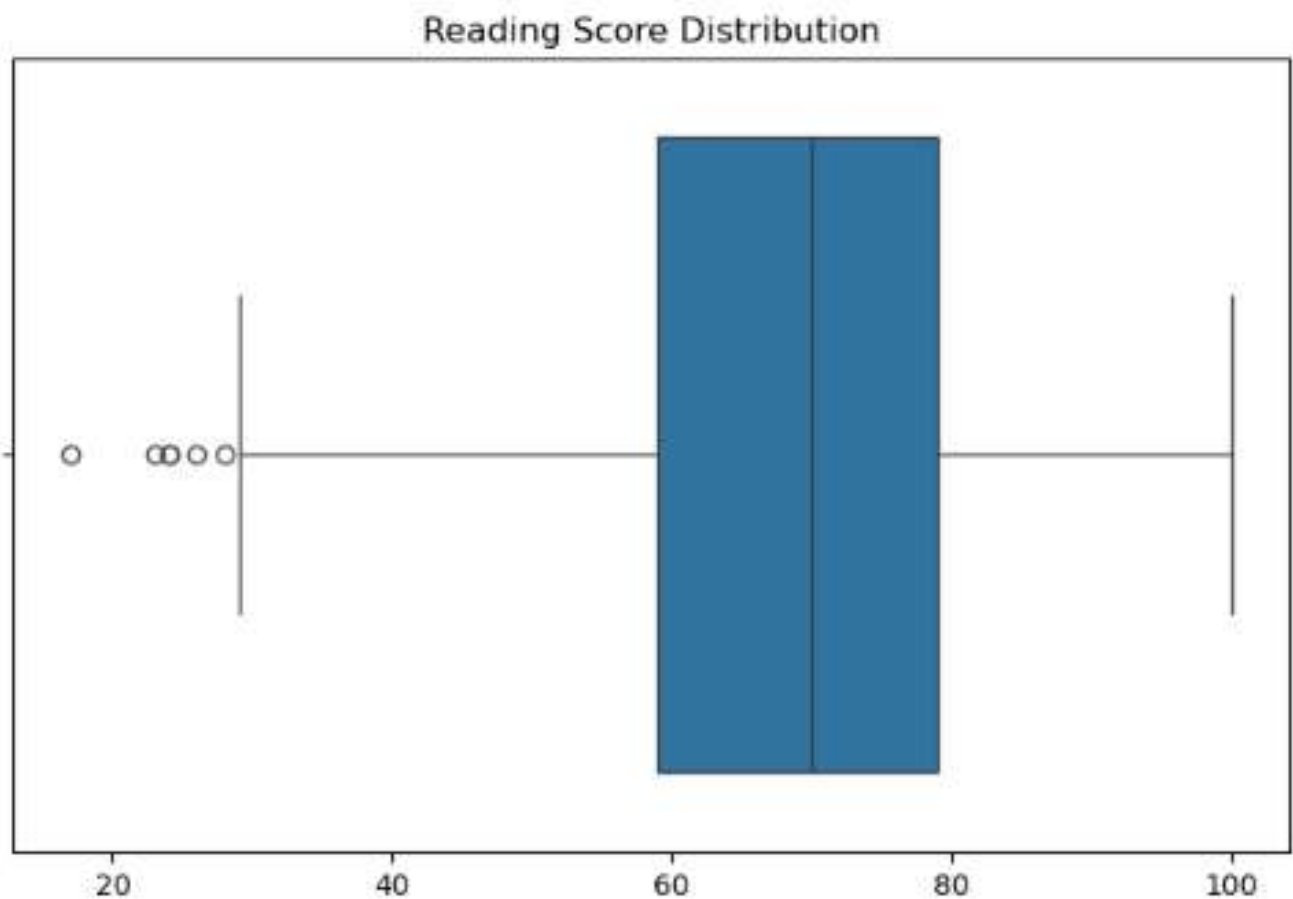
5. Reading Score Distribution

```
[25]: plt.figure(figsize=(8,5))

sns.boxplot(
    x=df["reading score"]
)

plt.title(
    "Reading Score Distribution"
)

plt.show()
```



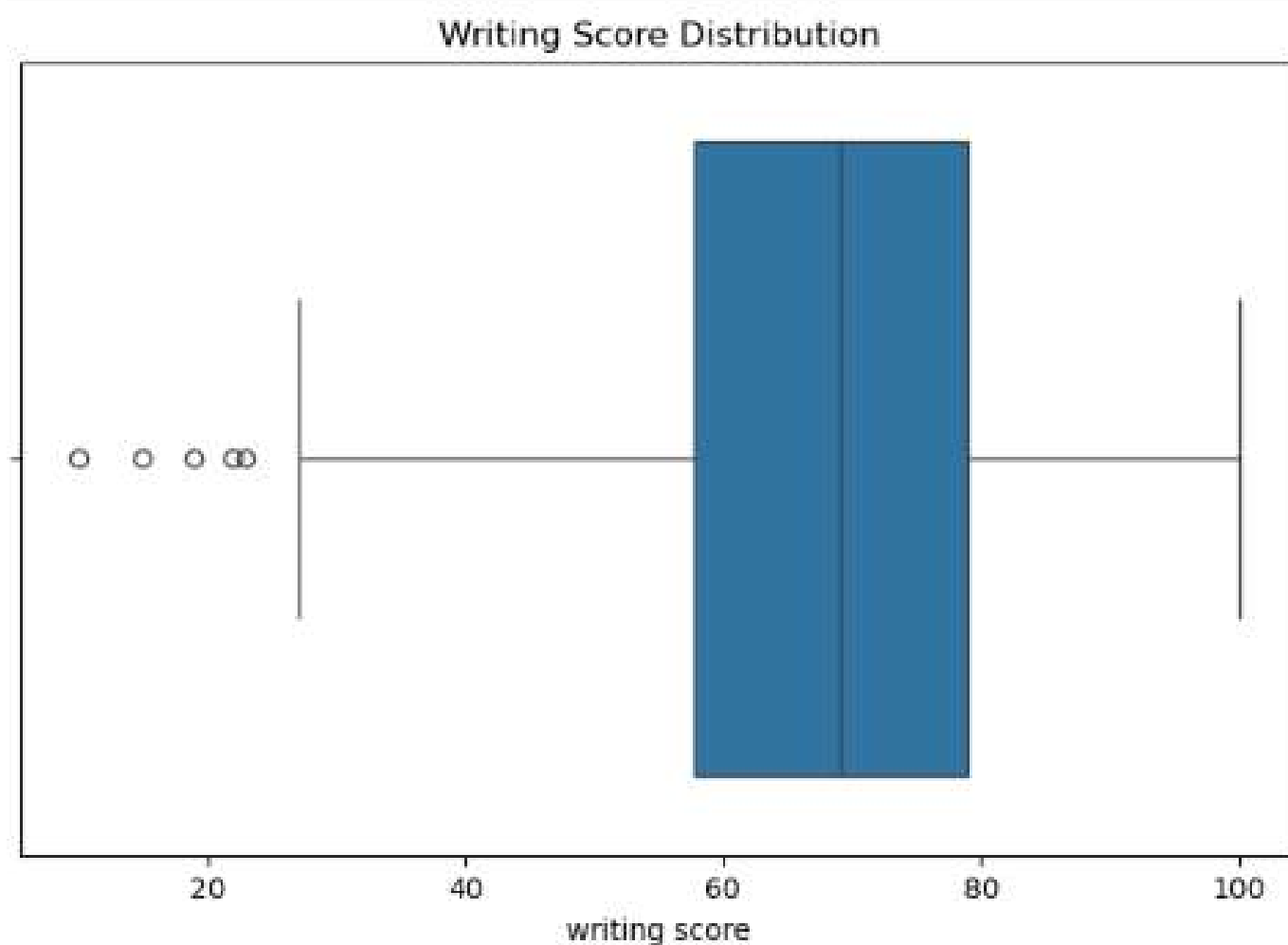
6. Writing Score Distribution

```
27]: plt.figure(figsize=(8,5))

sns.boxplot(
    x=df["writing score"]
)

plt.title(
    "Writing Score Distribution"
)

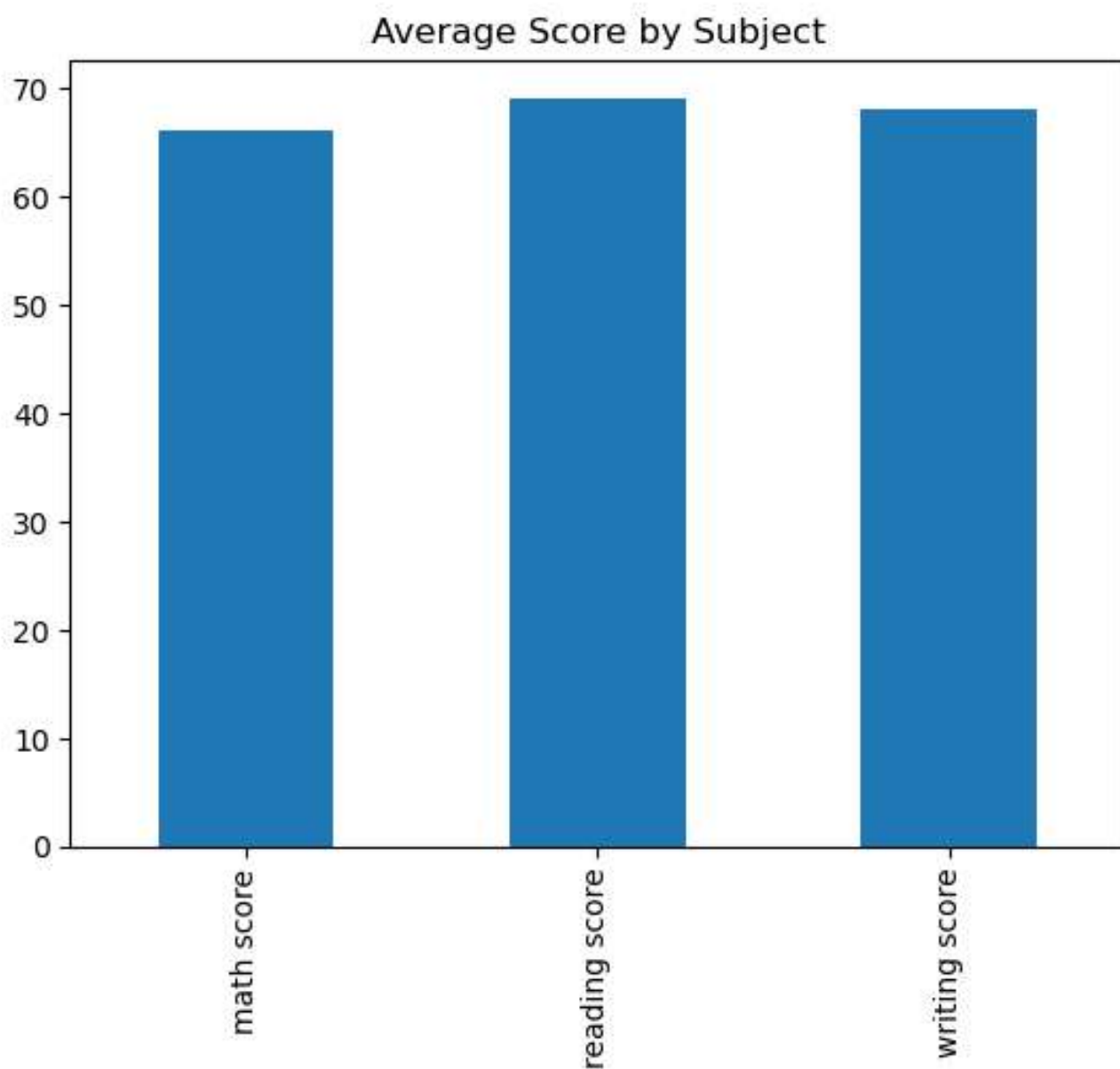
plt.show()
```



1. Subject-wise Performance Comparison

```
9]: subject_avg = df[["math score", "reading score", "writing score"]].mean()

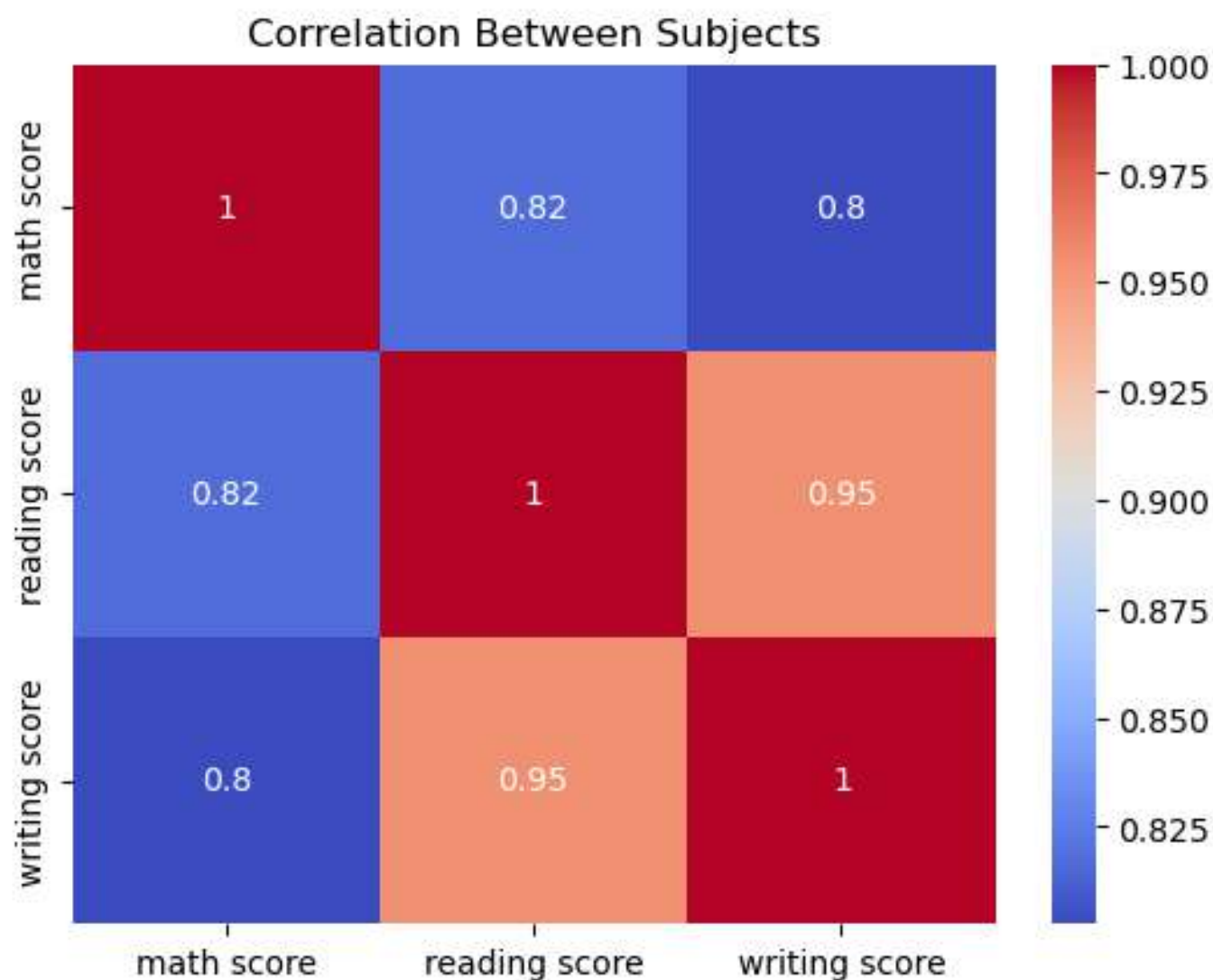
subject_avg.plot(kind="bar")
plt.title("Average Score by Subject")
plt.show()
```



2. Correlation Heatmap

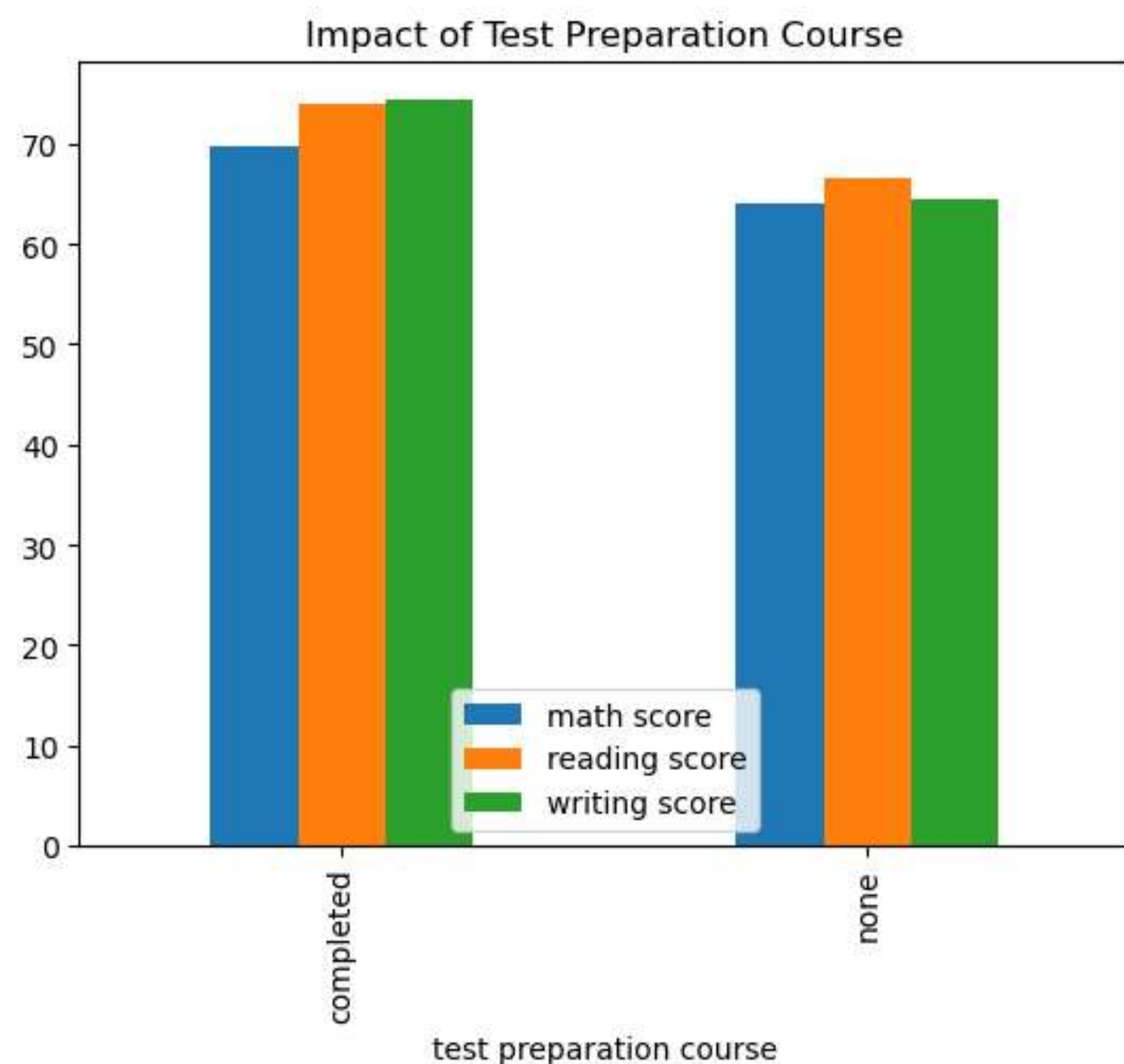
Shows relationships between scores.

```
sns.heatmap(  
    df[["math score", "reading score", "writing score"]].corr(),  
    annot=True,  
    cmap="coolwarm"  
)  
plt.title("Correlation Between Subjects")  
plt.show()
```



3. Test Preparation Course Impact

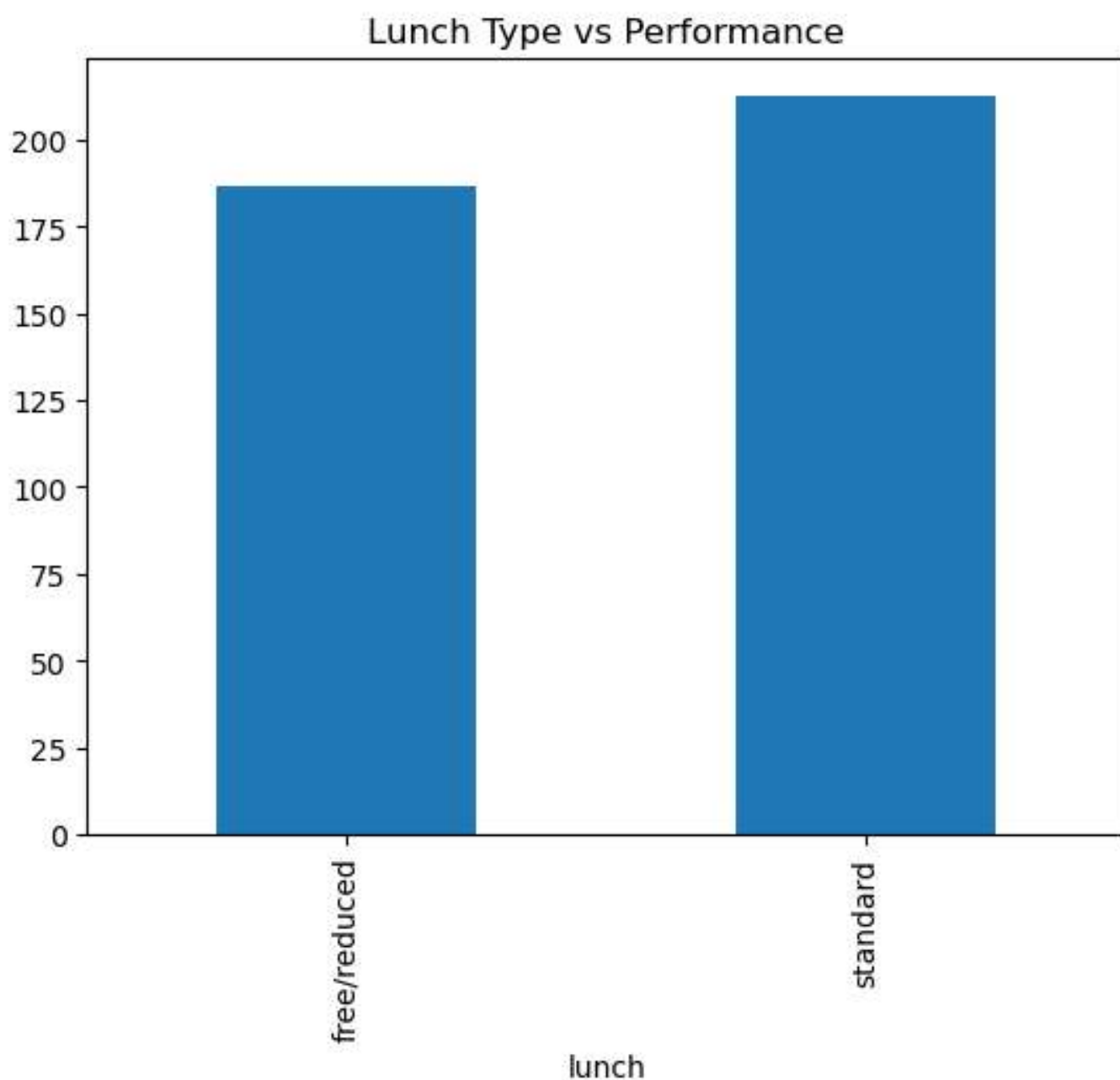
```
prep = df.groupby(  
    "test preparation course"  
)[["math score", "reading score", "writing score"]].mean()  
  
prep.plot(kind="bar")  
plt.title("Impact of Test Preparation Course")  
plt.show()
```



4. Lunch Type vs Performance

```
lunch_scores = df.groupby("lunch")["total_score"].mean()

lunch_scores.plot(kind="bar")
plt.title("Lunch Type vs Performance")
plt.show()
```

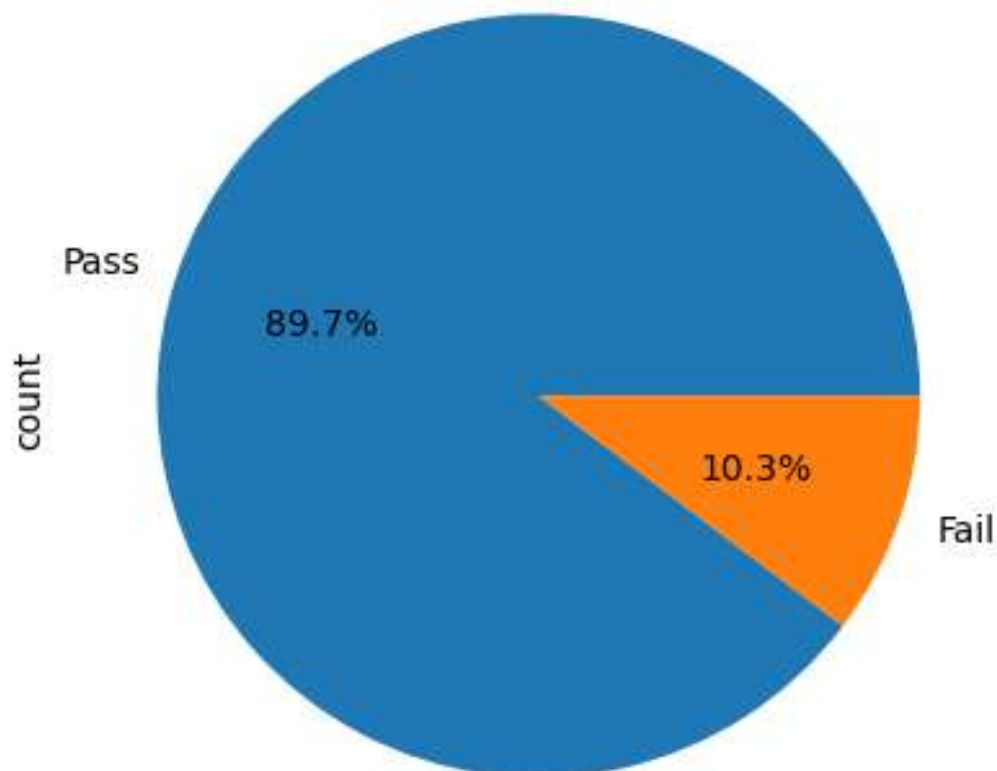


6. Pass/Fail Analysis

Create a new column:

```
df["Result"] = df["total_score"].apply(  
    lambda x: "Pass" if x >= 150 else "Fail"  
)  
  
df["Result"].value_counts().plot(kind="pie", autopct="%1.1f%%")  
plt.title("Pass vs Fail")  
plt.show()
```

Pass vs Fail



7. Performance Categories

```
def grade(score):  
    if score >= 250:  
        return "Excellent"  
    elif score >= 200:  
        return "Good"  
    elif score >= 150:  
        return "Average"  
    else:  
        return "Poor"  
  
df["Performance"] = df["total_score"].apply(grade)
```

```
sns.countplot(x="Performance", data=df)  
plt.title("Student Performance Categories")  
plt.show()
```

